
Lab 06

Data Entry, Transformations & Univariate Stats

Instructions

This worksheet will walk us through how to enter data into SPSS, how to transform/recode that data as needed, and run some basic univariate statistics.

Looking Forward: These basic SPSS skills will be needed for all of the future SPSS assignments, especially the final project. It will be important to have a firm grasp of the recode/select/transform operations, as your final projects will require you to apply these in a unique way.

Walk-through (10 points)

The dataset (shown below) includes the following variables (with values & labels for qualitative variables):

- **Unique number for each actor in dataset** (num)
- **Year of release** (year)
- **Type of media** (type) — Values: 1 = Film, 2 = TV Show, 3 = Serial
- **Height of Clark Kent/Superman actor in meters** (clark_height)
- **Height of Lois Lane actor in meters** (lois_height)

Things to do:

1. Using the **Variable View** specify the attributes of each variable (be sure to use Labels and Values).
2. Using the **Data View** enter the data.

Table 1

num	year	type	clark_height	lois_height
1	2025	1	1.93	1.60
2	1978	1	1.93	1.72
3	2001	2	1.90	1.71
4	2006	1	1.89	1.65
5	1951	1	1.86	1.63
6	2013	1	1.85	1.63
7	1948	3	1.85	1.62
8	2021	2	1.82	1.68
9	1993	2	1.81	1.68
10	1988	2	1.83	-99.00
11	1989	2	1.83	-99.00

Do each of the following transformations – one at a time!!

- Be sure to check if you got what you want by looking at the variable view & running a frequencies
 - Be sure to make Labels and Values for each (mind-numbing at the time, but important later...)
3. Make two new variables called `clark_height_in` and `lois_height_in` that reflect the height of each actor in inches rather than meters. Do this by computing the values of `clark_height` and `lois_height` from meters to inches. (*HINT: 1 meter = 39.37 inches!*)
 4. Make a variable called `clark_grp` by recoding the `clark_height_in` variable into a new variable (we never want to recode into the original variable as it overwrites our data!). This new variable will group Clark Kent/Superman actors into those who are: 1 = under 6ft tall, 2 = 6ft or taller. (*HINT: 1 foot = 12 inches*)
 5. Make a variable called `height_diff` that reflects the height difference between Clark Kent and Lois Lane actors in inches. Do this by computing the difference (`height_diff = clark_height_in - lois_height_in`).
 6. Make a variable called `height_gap` by recoding `height_diff` into a new variable. This will categorize the height difference into three categories: 1 = minimal (< 6 inches), 2 = average (6-8 inches), and 3 = big (> 8 inches).
 7. Select only those cases where the actors debuted in media that was released after 1975 (using `year`) to include in your analyses. (*HINT: After selecting cases, you should see some rows with a diagonal line through the row number in Data View, indicating they are not selected. The analyses you run will only include the selected cases.*)
 8. Get the mean, std, and SEM for all variables (entered and created) and enter those on the homework checker. The checker will also ask you whether the mean for each variable can be meaningfully interpreted.
 9. Grab a screenshot of the homework checker screen, showing all of your input answers in green (correct). Paste it at the end of the take-home section. **Your TA will not grade the assignment without this.**
 10. Following the example on the handout, and using the template below, make a “Table 1” for the following variables: `height_diff`, `clark_grp`, and `height_gap`

**Table 1.
Table 2**

Variable	M(SD)	N(%)
height_diff	—	—
clark_grp		
6ft or taller		—
Under 6ft		—
height_gap		
Average		—
Big		—
Minimal		—

Note. See APA manual beginning on p. 210 for additional examples.

11. **Save the dataset (and syntax windows)! You will use it in the next section! NOTE: Before you do, return to the “Select Cases” window and click the “All Cases” radio button to return to analyzing all the data!**

Take-Home (20 points)

Instructions

In this section, you will expand the dataset you created in the previous section by adding new ratings variables. You will use two publicly available sources of data: Rotten Tomatoes (an entertainment review site) and Letterboxd (social media site that lets users rate films). First, open your previously saved dataset, then add the following new variables:

- **Rotten Tomatoes critics score (0-100% scale)** (rt_critics_score)
- **rt_critics_count** (rt_critics_count)
- **Rotten Tomatoes audience score (0-100% scale)** (rt_audience_score)
- **Number of Rotten Tomatoes audience reviews** (rt_audience_count)
- **Total number of users that liked the film on Letterboxd** (ldb_likes)
- **Letterboxd users average rating (1-5 stars)** (ldb_scores)

Things to do:

1. Using the **Variable View** specify the attributes of each variable (be sure to use Labels and Values).
2. Using the **Data View** enter the data.

Table 3

rt_critics_score	rt_audience_score	ldb_likes	ldb_scores
83	90	1,105,511	3.9
88	86	99,115	3.7
78	72	-99	-99.0
72	60	26,076	2.7
-99	79	744	2.6
57	75	204,463	3.0
83	90	-99	-99.0
88	84	-99	-99.0
86	86	-99	-99.0
-99	-99	-99	-99.0
-99	-99	-99	-99.0

- **Do each of the following transformations – one at a time!!**
 - Be sure to check if you got what you want by looking at the variable view & running a frequencies
 - Be sure to make Labels and Values for each (mind-numbing at the time, but important later...)
- 3. Make a new variable called `tomatometer` by recoding `rt_critics_score` using the Rotten Tomatoes' classification standards. We want this to represent whether the media was liked by at least 60% of critics. Let the categories be: 1 = Rotten; and 2 = Fresh.
- 4. Make a variable called `rt_avg` an average of the number of critics and audience members that liked the media on Rotten Tomatoes (using `rt_critics_score` and `rt_audience_score`).
- 5. Make a variable called `rt_diff` that reflects the weighted difference between the number of critics and audience members that liked the media on Rotten Tomatoes. Compute it as $((rt_critics_score \times rt_critics_count) - (rt_audience_score \times rt_audience_count)) / (rt_critics_count + rt_audience_count)$.
- 6. Make a variable called `popular` by recoding `ldb_likes` into a new variable. This will categorize the popularity on Letterboxd into three categories: 1 = low (< 1,000 likes), 2 = mid (1,000-100,000 likes), 3 = high (> 100,000 likes).
- 7. Select only those cases where there are at least 5 critics scores (using `rt_critics_count`) to include in your analyses.
- 8. Get the mean, std, and SEM for all the **new variables** (entered and created) and enter those on the homework checker. The checker will also ask you whether the mean, std, and sem for each variable can be meaningfully interpreted.
- 9. Grab a screenshot of the homework checker screen, showing all of your input answers in green (correct). Paste it at the end of this section. **Your TA will not grade the assignment without this.**
- 10. Following the example on the handout, and the template you used above, make a "Table 2" for the following variables on the next page: `tomatometer`, `rt_avg`, `rt_diff`, and `popular`

Table 2.
Table 4

Variable	M(SD)	N(%)
rt_avg	—	—
rt_diff	—	—
tomatometer		

Variable	M(SD)	N(%)
Fresh		—
Rotten		—
popular		
High		—
Low		—
Mid		—

Note. M = Mean; SD = Standard Deviation; N = Sample Size.

11. **Save the dataset (and syntax windows)! You will need this dataset next week! NOTE:**
Before you do, return to the “Select Cases” window and click the “All Cases” radio button to return to analyzing all the data!

Homework Checkers

Paste Homework Checkers screenshots here: